Models, Implementations and Inferences

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Contents

[Arun: This document is nowhere near finished - it will be improved upon and updated as time goes on with more and more focused models]

1 Integrate and Fire Neurons

Integrate and fire (or Leaky I&F) neuron model is the simplest bottom-up rate code model that can be considered. "Bottom-up" models are a class of dynamical modelling systems designed to simulate neuron firing based on the physiological and biological structure of the neuron itself.

Neurons generate electrical pulses called **action potentials** that travels enough distance to influence the firing or inhibition of one or more neighboring neuron dendrites. These neurons are usually in complicated tree structures thereby allowing each neuron to receive and give out information and inputs to and from many neurons hence forming networks. These inputs and outputs are electrical pulses transfered through the neuronal membrane whose electrical potentials is usually negative by default. It can either become more negative, called **hyper-polarization** or get more positive, even all the way towards 0 mV, which is called **de-polarization**. With sufficient de-polarization a positive feedback loop is created inside the neuron physiology and a high, net 100 mV potential spike that lasts for a extremely short time in the membrane is fired and this propagates through the network thereby influencing other neurons along the way. This spike fired is often called as the action potential.

After an action potential is discharged, it becomes impossible to produce another firing for a period of time, this period is called as the **refractory period**. **Absolute refractory period** lasts a few milliseconds and **relative refractory period** lasts a longer period of tens of milliseconds.

This phenomenon can be dynamically modeled by assuming that the neuron fires an action potential when its membrane potential reaches a threshold value of around -55 mV to -50 mV denoted by V_{th}. In the model the firing is done when V_{th} is reached and the membrane potential resets to a much lower value such that $V_{reset} < V_{th}$. In the simplest version, the entire membrane conductance is modeled as a single leakage term of the form $g_L(V - E_L)$. This model mathematically resembles an RC circuit and obeys an ODE of the form,

$$\tau_m \frac{dV}{dt} = E_L - V + R_m I_e \tag{1}$$

where E_L is often associated with the resting voltage V_{rest}, R_m is the membrane resistance and I_E denotes the driving current through the membrane.

The behavior of the neuron satisfying the above equation is given in 1 where the input driving current I_e with a 2 nA pulse from 100 ms to 400 ms.

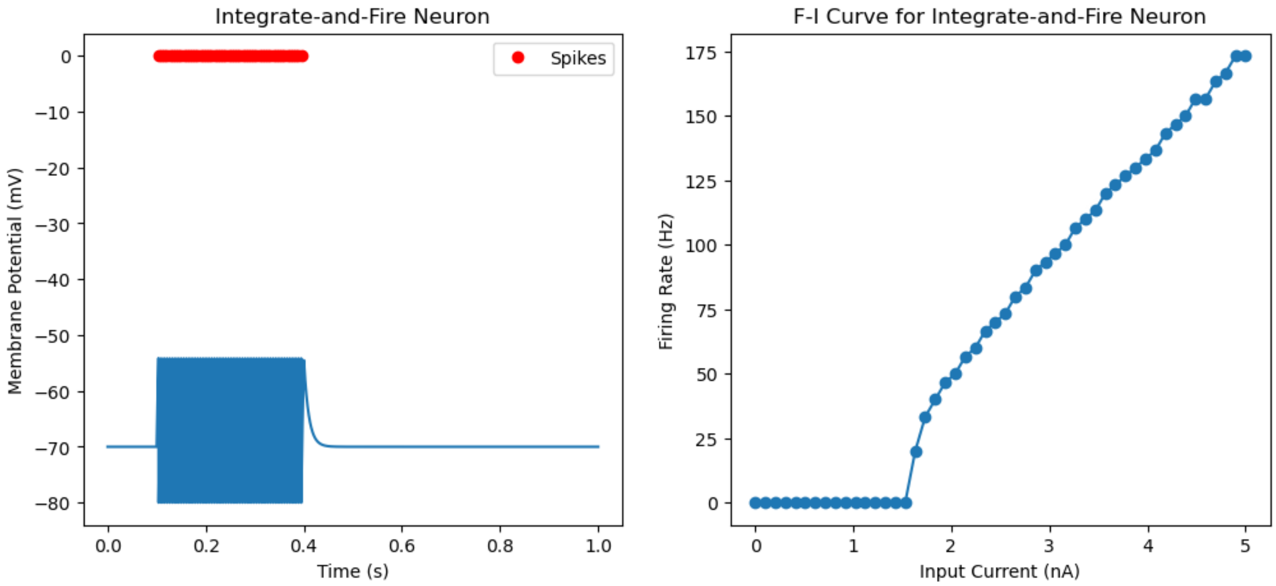

Figure 1: The values of constants, voltages and resistances chosen for this simulation are, $V_{rest} = -70$ mV, $V_{reset} = -80$ mV, $V_{th} = -54$ mV, $R_m = 10$ MΩ, and $\tau_m = 10$ ms. (Left) Shows the neuron firing when driving current is turned on between 100 ms and 400 ms. (Right) Shows the firing rate (Hz) with respect to the input current (nA)

The right panel in 1 shows how the firing rate, defined by

$$f = \frac{N_{spikes}}{T} \tag{2}$$

changes when the input current is varied. Based on the equation given in 1, a spike is triggered when the steady state voltage V_∞ is greater than the threshold voltage, where $V_\infty = V_{rest} + R_m I_e$. This can happen only when,

$$I_e > \frac{V_{th} - V_{rest}}{R_m}$$

For the system considered with the voltages and resistance values mentioned in 1, the value of I_e for which the steady state voltage exceeds the threshold occurs only when $I_e \geq 1.6$ nA which is clearly seen to be the case as shown in the right panel of the figure.

The figure also shows that the neuron fires without any damping during the time the driving current is supplied which is a very simplistic case. This can be improved by including a rate adaptation.

This spike rate adaptation which delays the next subsequent firing by a neuron can be introduced by adding a coupling term to the original ODE, which in turn satisfies another simple equation in the following way,

$$\begin{aligned} \tau_m \frac{dV}{dt} &= E_L - V + R_m I_e - A_r(V - E_K) \\ \tau_s \frac{dA_r}{dt} &= -A_r \end{aligned} \tag{3}$$

This increases the **interspike intervals** everytime firing occurs thereby slightly delaying the next spike as shown in the figure below.

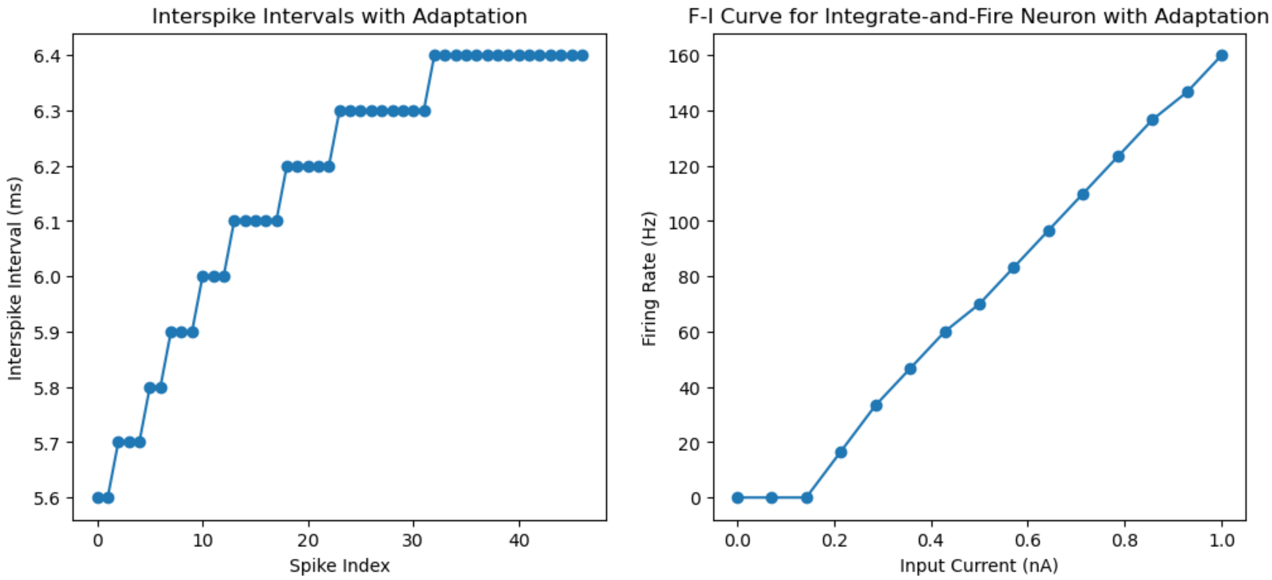

Figure 2: Integrate and fire neuron system with spike rate adaptation included in Eq.3 with values $V_{rest} = -65$ mV, $V_{reset} = -65$ mV, $V_{th} = -50$ mV, $R_m = 90$ MΩ, $\tau_m = 30$ ms, $\Delta A_r = 0.06$, $E_k = -70$ mV and $\tau_s = 100$ ms. (Left) Shows the increase in the interspike intervals as a function of the spike index. (Right) Shows the firing rate as a function of the input current.

This model can further be built upon by considering presynaptic spikes received from neighboring neuron(s). This is done by adding an excitatory synaptic conductance term to the original model. For this model we assume that the probability of release on receipt of a presynaptic spike is 1. This presynaptic spike term P_s is further assumed to obey a set of dynamical equations with a separate damping term. To incorporate multiple presynaptic spikes into the model, the full system of equations to be solved becomes,

$$\begin{aligned}
\tau_m \frac{dV}{dt} &= E_L - V(t) + R_m I_e - A_r P_s(t)(V(t) - E_s) \\
\tau_s \frac{dP_s}{dt} &= eP_{max} z(t) - P_s(t) \\
\tau_s \frac{dz}{dt} &= -z(t)
\end{aligned} \tag{4}$$

where $z(t)$ is hardset to 1 whenever a presynaptic spike arrives thereby damping the next spike production. This model for presynaptic spikes produces an α function response since the equation obeyed by P_s solves to give $P_s(t) = P_{max}\,(t/\tau_s)\,e^{1-t/\tau_s}$. On triggering synaptic events at fixed times at $[50, 150, 190, 300, 320, 400, 410]$ ms, the following behavior is observed.

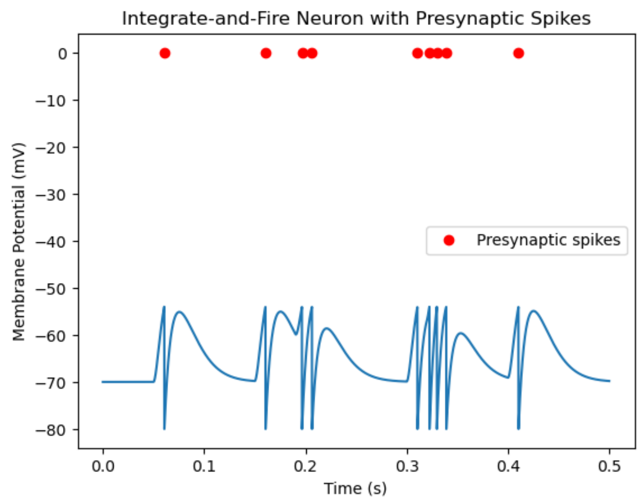

Figure 3: Voltage diagram for triggered presynaptic spikes at fixed times for the system values described in 1 with additional constants in Eq.4 given by $E_s = 0$ V, $I_e = 0$ nA, $\tau_s = 10$ ms and $A_r = 0.5$.

We see that the presynaptic spike input acts not just as a current but as a separate conductance term that pulls the voltage towards 0. We can look at the induced synaptic current and the total conductance introduced by the presynaptic spikes in the figure below.

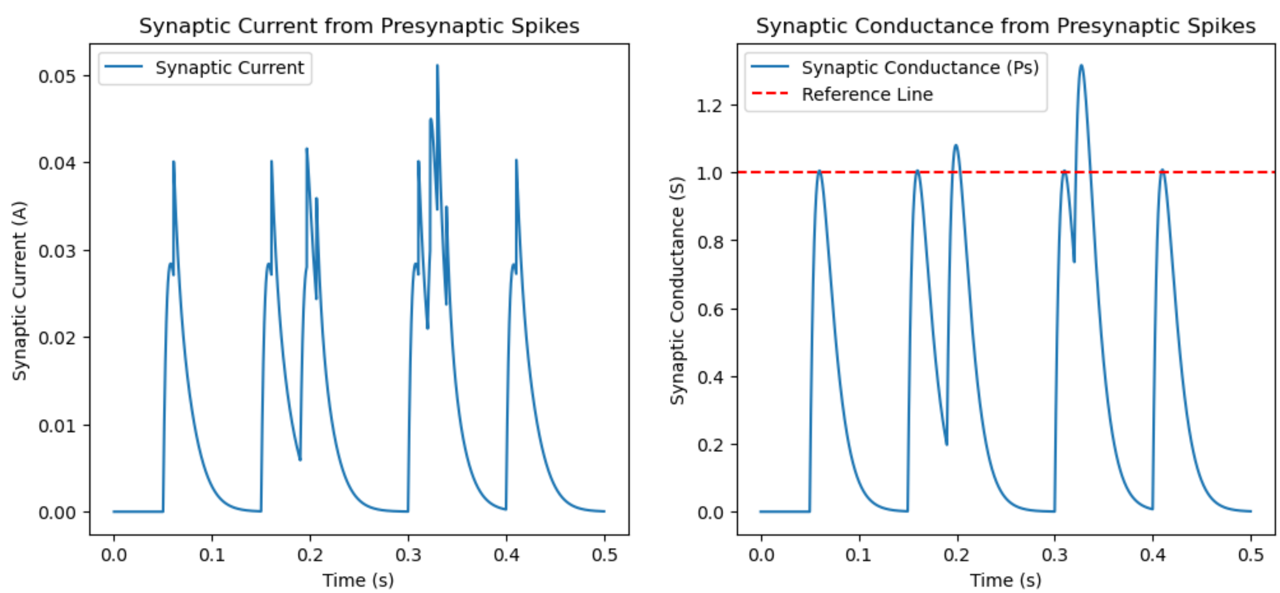

Figure 4: Current and conductance spikes introduced by the presynaptic inputs. It can be seen that the synaptic conductance exceeds 1 at high frequency inputs

To stop P_s from exceeding P_{max} when multiple presynaptic spikes are fired at high frequency, we can introduce an upper bound by modifying the first term of the respective equation into $eP_{max}(1 - P_s)z(t)$.

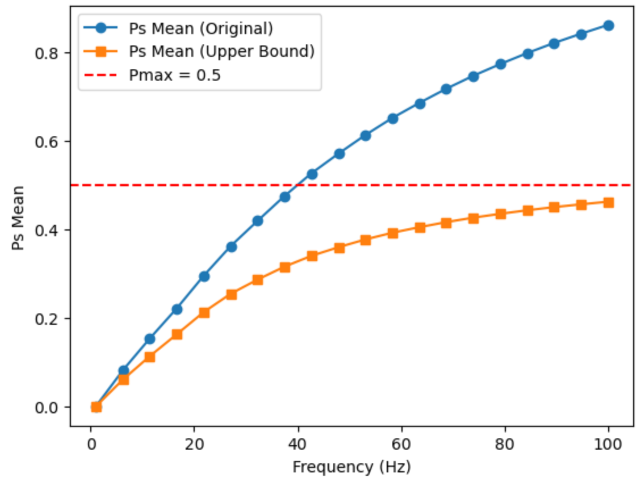

Figure 5: Trial averaged P_s values with (orange) and without (blue) modification shows that the modified system remains bounded within the maximum value with $P_{max} = 0.5$.

1.1 Inferences

- The leaky integrate-and-fire neuron provides the simplest dynamical model capable of converting continuous input currents into discrete spike trains.
- A neuron behaves as a leaky temporal integrator: incoming currents accumulate in the membrane potential while leakage continuously drives the voltage back towards the resting state.
- Spiking occurs only when the steady-state membrane voltage exceeds the firing threshold. This introduces a minimum current required for sustained firing, explaining the existence of a threshold current observed in the F-I curve.
- Above threshold, increasing input current leads to an increase in firing frequency, demonstrating how neurons can encode stimulus intensity through firing rate.
- The basic LIF model produces unrealistically regular spike trains under constant stimulation because it lacks mechanisms that reduce excitability after repeated firing.
- Spike-rate adaptation introduces a slow negative feedback process that increases interspike intervals and lowers the effective firing rate over time. This provides a simple mechanism for gain control and prevents excessive firing.
- Synaptic interactions are more naturally represented as conductance changes rather than direct current injections. Presynaptic spikes transiently modify membrane conductance and drive the voltage towards a synaptic reversal potential.
- The α-function synapse introduces a finite rise and decay time for synaptic events, making synaptic transmission a dynamical process with memory rather than an instantaneous event.
- Multiple closely spaced presynaptic spikes can accumulate and produce larger conductance changes, illustrating temporal summation of synaptic inputs.

- Introducing an upper bound on synaptic conductance prevents unphysical growth at high firing rates and better reflects the finite availability of synaptic resources.

Overall, the progression of the models implemented in this section shows how increasingly realistic neuronal behaviour can emerge from adding a small number of additional dynamical variables while retaining computational simplicity.

2 Excitatory - Inhibitory coupled neuron model

2.1 Context

In multi-neuron models it becomes necessary to track both the pre- and post-synaptic firing rates which will be denoted by u and v respectively. This becomes necessary since in networks, there can be three types of synaptic inputs namely,

- **Feed-forward**: brings input to a region from another region located at an earlier stage in a processing pathway.
- **Recurrent**: interconnect neurons within a particular region along the same stage in the processing pathway.
- **Top-down**: sends signals back from areas located at later stages in the processing pathway.

It is better to focus mainly on firing rates to track networks since it is much simpler to simulate and compute, along with the following advantages,

- rates avoids short time scale dynamics involved in spike statistics ,
- less number of free parameters compared to spiking models ,
- avoids the general randomness/stochasticity of spike firing (stochastic firing can also be modelled by firing rate codes if necessary),
- many neurons can be bundled into units based on their average output of firing rate which can then be used to build networks.

This method involves replacing the input spike density with the presynaptic firing rate as the initial condition. Replacing the spike density with the firing rate is justified if the quantities of relevance in the system are relatively insensitive to trial to trial fluctuations in spike sequences. This presynaptic firing rate is then taken to be the input of the neurons in the network. Basically, if the presynaptic spike train inputs are uncorrelated (no synchronous firing) then presynaptic firing rates can be used and vice versa.

As we saw in the previous section, firing rate response curves are determined the injected driving current. Thus total synaptic input $\approx$ total injected synaptic current I_s which we can use to compute the post synaptic firing rate.

The dynamical equation for such a model is given by,

$$\tau_s \frac{dI_s}{dt} = -I_s + \vec{w} \cdot \vec{u} \tag{5}$$

where $\vec{w}$ is the vector of synaptic weights for the inputs and $\vec{u}$ is the vector of presynaptic firing rate inputs. The amplitude and sign of the presynaptic weights determines if the input is **excitatory** or **inhibitory** with

$w_b > 0$ and $w_b < 0$ respectively and the subscript b is the input index number with range from $[1, ..., N_u]$ where N_u is the total number of input rates.

For a constant synaptic current I_s, the post-synaptic firing rate, v, is given by $v = F[I_s]$, where F is called as the **activation function**. To build models two types of activation functions are considered namely,

- Threshold linear function $F[I_s] = [I_s - \gamma]_+$, and
- Non-linear sigmoidal activation function.

So for time independent inputs $\vec{u}$, the constant current and the post-synaptic steady state firing rate is given by

$$\begin{aligned} I_s &= \vec{w} \cdot \vec{u} \\ v_\infty &= F[I_s] \end{aligned} \tag{6}$$

The above equations describe the full system completely.

For time dependent inputs, I_s becomes dynamic and the system of equations become,

$$\tau_s \frac{dI_s}{dt} = -I_s + \vec{w} \cdot \vec{u}, \text{ with} \tag{7}$$

$$v = F[I_s]$$

Another case, which is much more interesting occurs when the firing rate does not follow changes in I_s in which case we can alternate between,

$$\tau_r \frac{dv}{dt} = -v + F[I_s(t)], \text{ and} \tag{8}$$

$$\tau_s \frac{dI_s}{dt} = -I_s + \vec{w} \cdot \vec{u} \tag{9}$$

If $\tau_r << \tau_s$ then $v(t)$ settles to an equilibrium point quickly and we can just set $v = F[I_s(t)]$ like before and use Eq.(7). Similarly if $\tau_s << \tau_r$ then I_s settles to steady state and we can use $I_s = \vec{w} \cdot \vec{u}$ and use,

$$\tau_r \frac{dv}{dt} = -v + F[\vec{w} \cdot \vec{u}] \tag{10}$$

But neither the equation for $\frac{dv}{dt}$ nor $\frac{dI_s}{dt}$ provides accurate description/prediction for the dynamics of the firing rate for all frequencies of injected currents thus making it necessary to develop a more complex model for a compound function $F(I_s, v)$.

2.2 Equations

When considering a network with feed-forward and recurrent inputs and feedbacks it is important to separate the influences clearly to avoid confusion.

Suppose the network has N_v units with output rates v_a with $a = 1, ..., N_v$ for a feed=forward model the firing rate equation becomes

$$\tau_r \frac{d\vec{v}}{dt} = -\vec{v} + F[\mathbf{W} \cdot \vec{u}] \tag{11}$$

where $\mathbf{W}$ denotes the synaptic weight matrix.

If the system also has recurrent inputs, then the input becomes a function of the post-synaptic firing rate as well which is moderated by another synaptic weight matrix $\mathbf{M}$.

These neurons can be excitatory or inhibitory on their post-synaptic targets as mentioned above, but **Dale's law states that a neuron cannot be excitatory to one target and inhibitory to another target** which simplifies the problem considerably.

To track these excitatory and inhibitory neurons separately we have to split the system giving us the final set of coupled equations for the E-I network model,

$$\begin{aligned}\tau_E \frac{d\vec{v_E}}{dt} &= -\vec{v_E} + F\left(\vec{h}_E + \mathbf{M}_{EE} \cdot \vec{v_E} + \mathbf{M}_{EI} \cdot \vec{v_I}\right) \\ \tau_I \frac{d\vec{v_I}}{dt} &= -\vec{v_I} + F\left(\vec{h}_I + \mathbf{M}_{IE} \cdot \vec{v_E} + \mathbf{M}_{II} \cdot \vec{v_I}\right)\end{aligned} \tag{12}$$

where the subscript E and I denote the excitatory and inhibitory populations, $\mathbf{M}_{AB}$ denotes the coupling synaptic weight matrix between the two populations and $\vec{h}_i = \mathbf{W} \cdot \vec{u}$ denotes the total feed-forward input.

2.3 Linear Model

Assuming all excitatory and inhibitory neurons to have a single firing rate v_E and v_I respectively and using a linear threshold activation function, the RHS of Eq. 12 becomes,

$$-v_i + [M_{AB}\ v_B + M_{BA}\ v_A - \gamma_i]_+ \tag{13}$$

where i, A and B can be E or I and the + denotes half-wave rectifier.

Such coupled systems could either evolve to settle into

- **fixed point**: points where both v_E and v_I becomes constant. Can be found by tracking the nullclines and finding intersection points,
- **limit cycle**: a closed orbit in phase plane into which the system spirals and settles if fixed point is complex and unstable.

Linear stability analysis can be performed around a fixed point by computing the Jacobian stability matrix and studying its eigenvalues. The dimension of the stability matrix depends on the number of coupled equations being solved, i.e number of distinct neuron populations considered in the model. The eigenvalues can be complex with the following stability conditions,

- If the real parts of both eigenvalues are negative, then the fixed point is stable
- If either of the real parts are positive, the point is unstable.
- If both eigenvalues are purely real, then the system moves exponentially towards the fixed point
- If they are complex conjugates of each other then if
  - real part is negative the system spiral towards the fixed point
  - real part is positive it spirals out away from the fixed point
- The imaginary part of the eigenvalue $\mathrm{Im}(\lambda)$ determines the frequency of oscillations near fixed points.

If a system transitions from a stable fixed point into a limit cycle a **bifurcation** has occured.

For a simple E-I 2 unit network, we solved the network equations to find the fixed point as we varied the time constant of the inhibitory equation τ_I. The values of constants considered for the model are $M_{EE} =$

1.25, $M_{II} = 0$, $M_{EI} = -1.0$, $M_{IE} = 1.0$ and $\gamma_E = -10$ Hz, $\gamma_I = 10$ Hz with $\tau_E = 10$ ms, τ_I was varied to study the system dynamics.

Thus this is a two variable system (r_E, r_I) being the two firing rates, with a connectivity matrix,

$$M = \begin{pmatrix} 1.25 & -1.0 \\ 1.0 & 0 \end{pmatrix}$$

whose phase portrait is given in Fig. 6

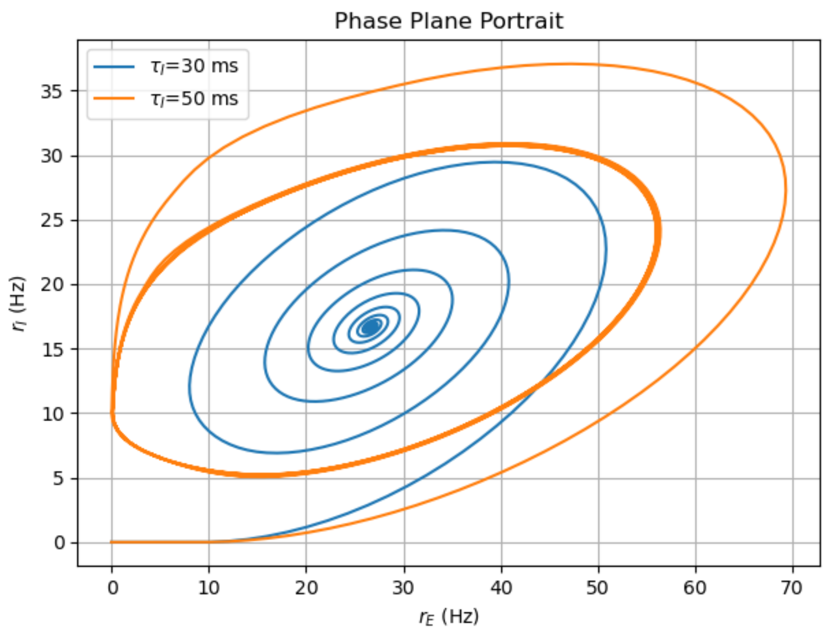

Figure 6: Phase portrait of the system described above for two values of $\tau_I = 30$ ms(blue) and 50 ms (orange). The system spirals and settles into the fixed point for stronger inhibition with $\tau_I = 30$ ms and it enters a stable limit cycle for weaker inhibition (orange)

We observe that for small time constant of inhibition, the perturbations decay and the system spirals to a fixed point. For a large inhibitory time constant, inhibition is delayed and excitation overshoots resulting in oscillations and the system settling into a stable limit cycle.

If we now track the eigenvalues of the stability matrix we see that the bifurcation occurs for $\tau_I \approx 40$ ms where the real part of the largest eigenvalue becomes positive. This can be seen in the figure below.

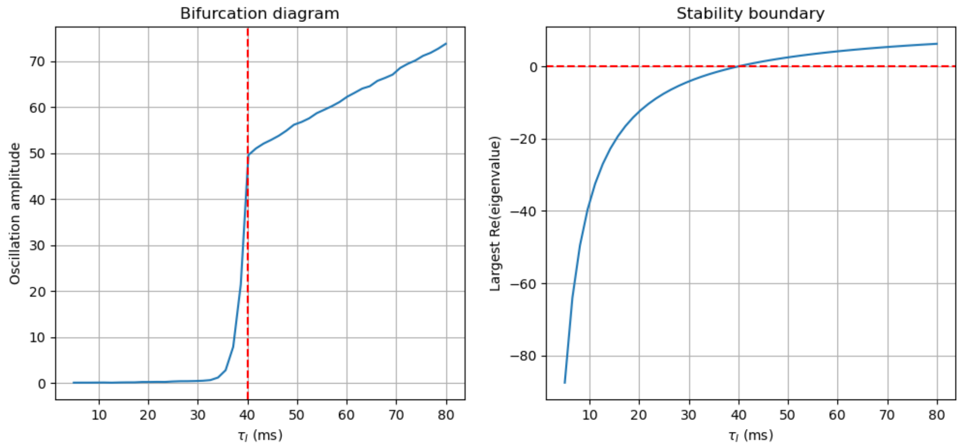

Figure 7: Bifurcation diagram showing the system transition from a stable fixed point into a stable limit cycle

Such a bifurcation is called a **Hopf bifurcation**. To summarize, we see that excitation tries to amplify activity and inhibition stabilizes activity. Slow inhibition creates delayed negative feedback and delayed negative feedback naturally produces oscillations thereby mimicking a neural rhythm.

2.4 Sigmoidal activation model

In this model the linear threshold activation function is replaced with a non-linear sigmoidal function of the form,

$$F(x) = \frac{100}{(1 + \exp(-\beta(x - x_0))} \tag{14}$$

This introduces another control parameter in the form of β which is the slope of the sigmoid function and also another possible result for the system to saturate to a maximum value. So τ_I controls the delay timescale and β controls the non-linear gain.

As a first result it is observed that there is no stable fixed point even in the previously safe value of $\tau_I < 40$ ms, with even a small $\beta = 0.15$ as seen in the figure below. This is understandable since the modified non-linear activation function completely destabilizes the system even for a small effective gain slope value.

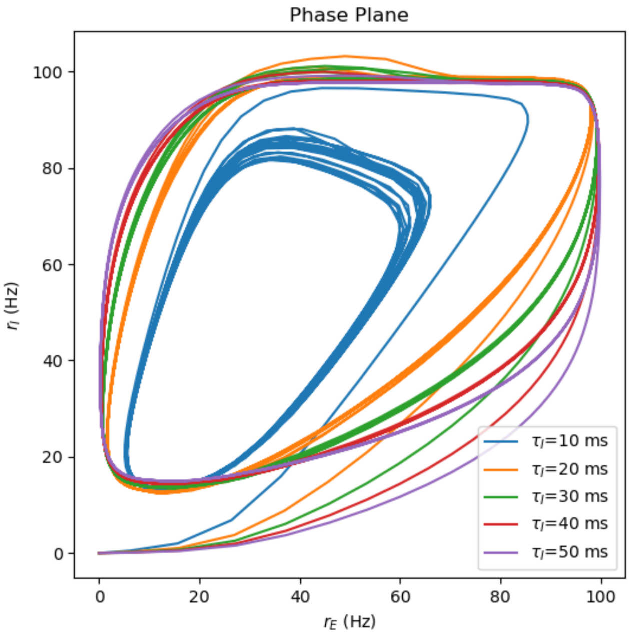

Figure 8: Only limit cycles are observed for $\beta = 0.15$ and various τ_I values

So the next step is to study the stability of the system as the two control parameters (τ_I, β) are simultaneously varied which is carried out for the same E-I system values as before in hopes of finding criteria for the existence of stable fixed points.

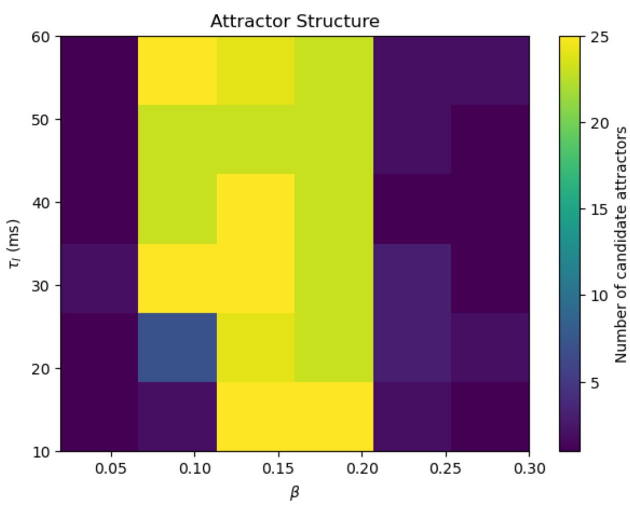

Figure 9: Plot showing the number of fixed points as a function of both τ_I and β. It can be seen that for low values of β there are stable fixed points for all τ_I and then the number of fixed points blow up to ≈ 26 before again collapsing to a single value.

As it can be seen from the contour plot Fig.9, inclusion of non-linearity makes the results more complicated. It can be observed that there exists stable fixed points for the system to settle into for small β and then the system destabilizes as β is increased. Even for $\beta = 0.10$ it can be seen that the system enters limit cycles for even small values of τ_I. This is recorded as multiple fixed points eventhough the system enters limit cycles because the code takes snapshots and limit cycles makes the trajectories end on different points on the same orbit depending on the initial condition and the simulation length. For high values of β, from $\beta \geq 0.20$, it can be seen that the number of fixed points again collapses to small numbers but upon closer inspection, it is observed that it is because the system saturates to a maximum value and stays there which is a new behavior. Interestingly, introduction of non-linear sigmoid activation stabilizes the system for small value of β in even the unstable bifurcated region of $\tau_I \geq 40$ ms as seen in the figure below.

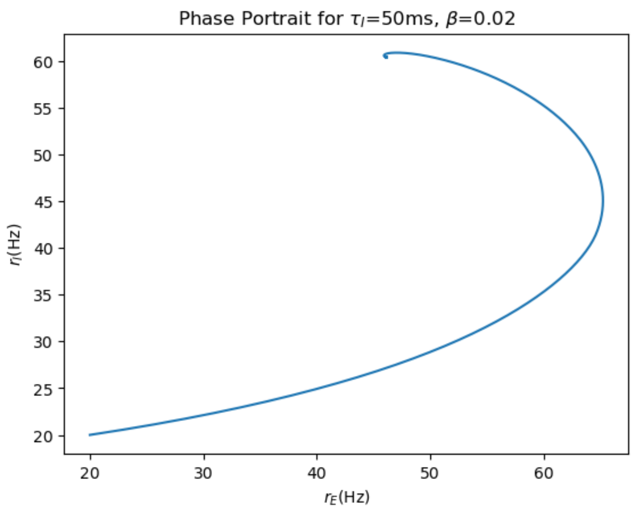

Figure 10: Stable collapse to a fixed point even in the previously unstable regime of $\tau_I = 50$ ms

Similarly we can also check that the high value regime of both control parameters is indeed saturation as seen in the image below.

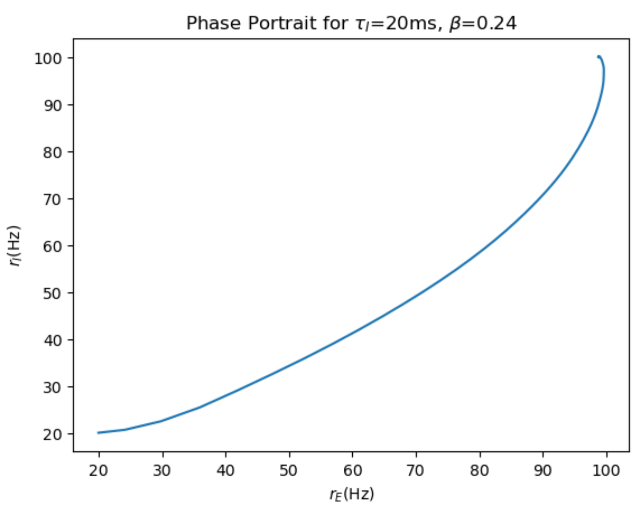

Figure 11: Saturation of the system to maximum value seen for previously stable $\tau_I = 20$ ms in the high β range

The observations can be reasoned as follows, as β is increased, the excitation dominates and pushes the system from stability to limit cycles and ultimately to a saturated attractor which is typical of a non-linear phase transition.

[Arun: Experimental bigplot]

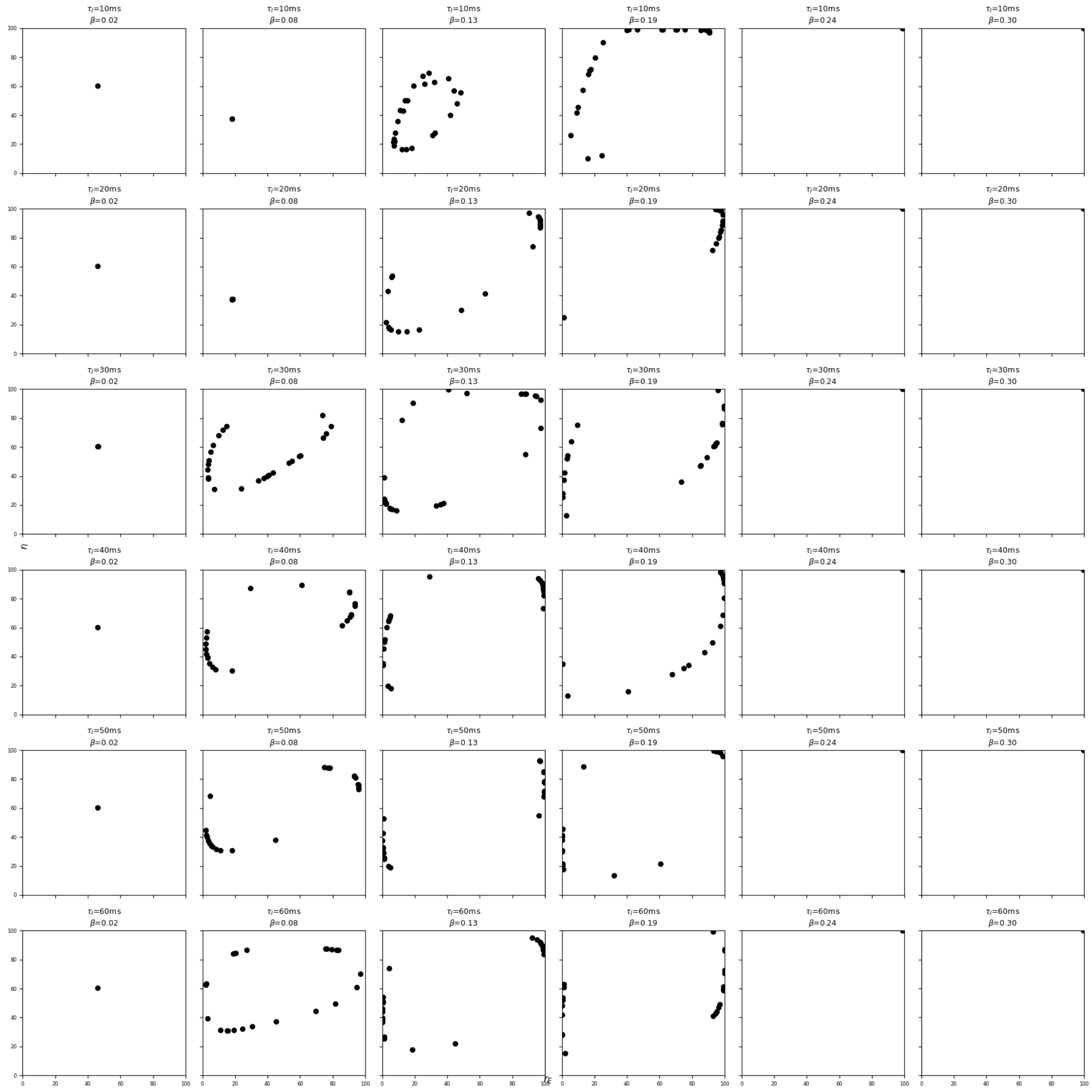

Figure 12: Full parameter snapshots of phase portraits showing the system devolve from stable fixed points to limit cycles to saturation. The last two columns are not empty there is a single saturated attractor point at $(100, 100)$

[Arun: Literature references to be added]

3 Hopfield Memory Network

Hopfield memory network is unique since this is a discrete time attractor computation in contrast to the continuous-time neural dynamics.

In typical long-term memory over periods of hours to years, the storage is done through the synaptic weight strength in the sense that the synaptic weights in a recurrently connected network are hard-set when a memory is stored. At a later time the network uses this as anchors to recreate the original pattern of activity that represents the stored memory. Persistent activity still occurs and is used to trigger the memory recall and to register the identity of the retrieved item but synaptic weights provide the long term storage.

This is called as associative network. A partial or approximate representation is used to recall the full memory. This is modeled dynamically by adjusting the recurrent synaptic weights in such a way that the network has a discrete set of fixed points identical to the stored memory patterns, i.e $\vec{v^m} = F(\mathbf{M} \cdot \vec{v^m})$ must be satisfied.

Hopfield networks are simple iteration rule systems that drive the system towards memories that are stored as attractors in the solution space. This is achieved by defining the system and the stored memory in a particular way. Traditionally defined as a recurrent associative memory networks that stores patterns and recovers them from noisy inputs to perform pattern completion. These models are not membrane physics but are memory retrieval and attractor relaxation performing collective computation in the simplest and the most abstract level. This is done by replacing $r_E(t)$ and $r_I(t)$ with a single post-synaptic firing rate denoted by $v_a(t) \in [+1, -1]$, where $a = 1, ...N$ denotes the different number of neurons in the memory and only its state is recorded as either ON (+1) or OFF (-1).

This is possible when time is discretized, the previously continuous differential equation of the form

$$\dot{r} + r = F(Mr + h)$$

becomes $r(t + \Delta t) \approx F(Mr(t))$. In Hopfield network this $F(Mr(t))$ is set to $\text{sgn}(Mr(t))$ or more accurately in this model the state $v_a(t)$ is iteratively updated as

$$v_a(t+1) = \text{sgn}\left(\sum_{a'} M_{aa'} v_{a'}\right) \tag{15}$$

This simply means that at next timestep neuron a becomes whatever the sign the total recurrent input tells it to be.

For the chosen model the weight matrix is defined to be

$$M_{aa'} = (1 - \delta_{aa'}) \sum_{m=1}^{P} v_a^m v_{a'}^m \tag{16}$$

This is called as a Hebbian memory rule with a simple system where suppose two neurons have the same sign in the stored memory, i.e $v_a^m v_{a'}^m = +1$ then the connection between them is positive and they reinforce each other and vice versa. The memory vectors v_a^m are simply just a binary pattern of +1 and -1 of length N.

When the system is initiated with a noisy corrupted input that mostly resembles the stored memory, the recurrent dynamics based on the iterator rule pushes the system towards the stored memory and cleans up the state. This is called as **attractor relaxation**.

This is checked by defining an overlap function,

$$q(t) = \frac{v(t) \cdot v^m}{N} \tag{17}$$

This function tracks the similarity such that if

- $q(t) = 1$: exact memory is restored,
- $q(t) = 0$: unrelated state
- $q(t) = -1$: inverse pattern is restored

Since the dynamics is symmetric under a sign flip both a memory v^1 and its inverse $-v^1$ satisfy the update rule. In this simple model it was easily seen that the rule chosen for the matrix restores the memory fully with the overlap function returning 1. We can explore the parameter space between P is the number of stored patterns and N is the number of neurons in each memory vector and for the update rule mentioned above, we obtain contour plot given in Fig.13.

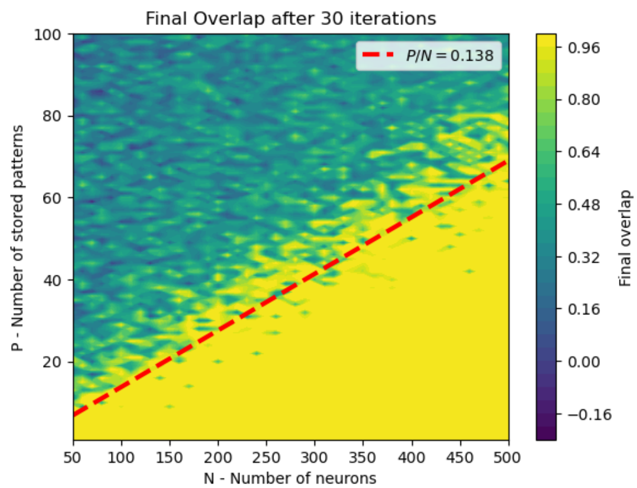

Figure 13: Contour of the overlap as a function of P and N for a 20% corrupted initial state. An approximate cut-off in efficiency is observed for $P/N \approx 0.14$.

We can also introduce another parameter called the storage capacity $\alpha = P/N$ which is more fundamental than individual P or N. We can now study how the overlap changes with α directly and see if we can reproduce the same cut-off observed in Fig.13. To do this, the overlap function is evaluated after 40 iterations for a range of α values with a control parameter of number of neurons N, in each stored memory. This gives us Fig.14

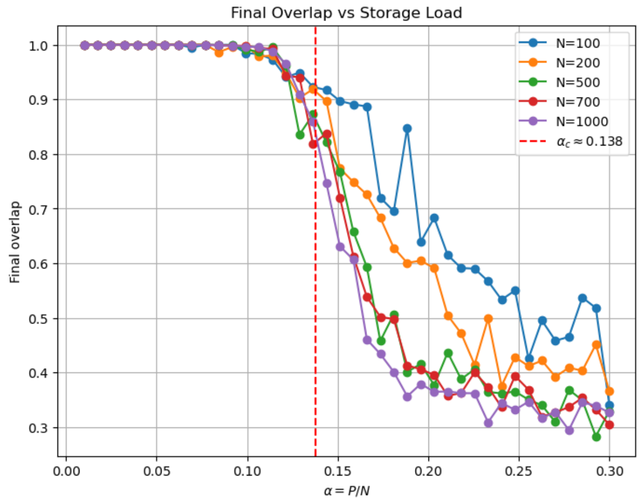

Figure 14: Memory retrieval efficiency drops drastically around $\alpha \approx 0.135$

This critical number of $\alpha \approx 0.138$ is not a numerical artifact but is a derivable result from mean field analysis of the system. The local field computed by each neuron has a form

$$h_i = \sum_j M_{ij} v^j$$

With our current form for the weight matrix,

$$M_{ij} = \sum_{m=1}^{P} v_i^m v_j^m, \text{ thus}$$

$$h_i = \sum_j \sum_{m=1}^{P} v_i^m v_j^m v^j, \text{which for a particular memory m = 1, on splitting the sum becomes}$$

$$h_i = \sum_j v_i^1 v_j^1 v_j^1 + \sum_{m \neq 1} \sum_j v_i^m v_j^m v_j^1 \tag{18}$$

The first term is the true memory while the second term is interference from other stored memories. The cross-talk from the other stored memories start dominating through the second term at this critical value of $\alpha \approx 0.138$ which is a unique result of the Hopfield memory network. This is reproduced to significant clarity in the results attached in the above figures of the section.

Having established the dependence on storage capacity, the next question is how robust memory retrieval remains when the initial state is increasingly corrupted. This can be studied by varying both the storage load (α) and the fraction of flipped bits in the initial condition.

This is done by analyzing the parameter space of α and the fraction of noise in the initial data within the range $[0.0, 0.5]$ which is no corruption to a state with half of its bits corrupted.

The overlap function for this chosen rule as a function of the flip fraction and α is given in Fig.15

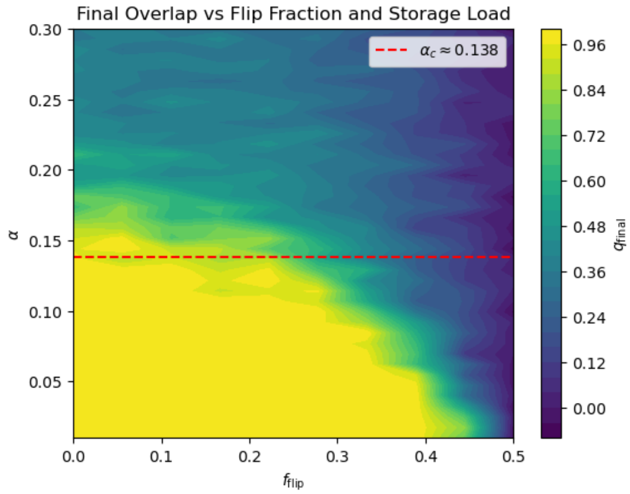

Figure 15: For the iteration rule defined above, the overlap function is plotted as a contour against the memory load α and the flip fraction. It is seen that Hopfield memory is capable of handling a significant amount of state corruption provided it has a low memory load.

The above figure reveals the interplay between two independent limitations of associative memory retrieval: storage capacity and input corruption. For low storage loads ($\alpha << \alpha_c$), the attractor basins remain large and the network successfully reconstructs memories even when a substantial fraction of the initial state is flipped. As the storage load approaches the critical capacity $\alpha_c \approx 0.138$, interference between stored memories reduces the basin size of each attractor, making retrieval increasingly sensitive to corruption. Beyond the critical capacity, memory interference dominates and reliable recall becomes impossible even for relatively clean initial states. The results therefore show that retrieval performance is determined not only by how corrupted the intial state is, but also by how densely memories are packed within the network.

3.1 Pseudo-inverse rule

The memory retrieval capacity of the Hopfield network can be drastically improved by altering the weight matrix. As an example altering the previously used weight matrix $M_{aa'}$ given in Eq.16 into,

$$M_{aa'} = (1 - \delta_{aa'}) \sum_{m,m'=1}^{P} v_a^m C_{mm'} v_{a'}^m \tag{19}$$

where $C^{-1}_{mm'} = \sum_{a=1}^{N} v_a^m v_a^{m'}$ is the inverse of the correlation matrix between stored memories. Inclusion of this inverse correlation matrix into the weight matrix definition removes the noise and cross-talk from the second term of Eq.(18) which should significantly increase the efficiency.

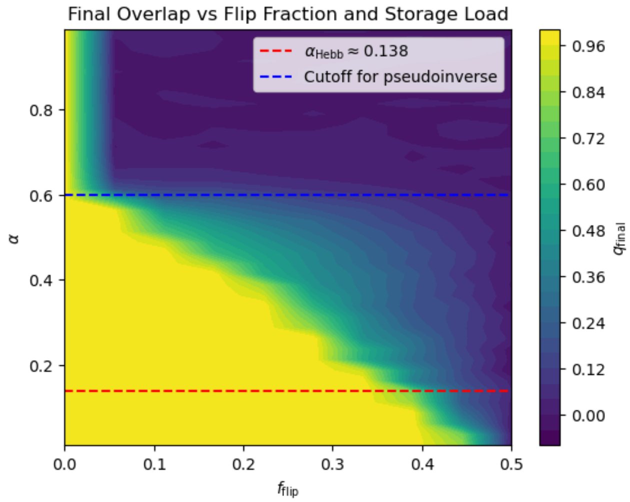

Figure 16: Contour of the overlap values after 30 iterations against the memory load α and the flip fraction based on the modified pseudo-inverse memory rule in Eq.19. The range of parameter scan for α is increased from 0.3 all the way to 1.0.

In the Hebbian memory rule, retrieval failed almost immediately once the system crossed the theoretical critical memory load capacity because of the fore-mentioned cross-talk noise. This pseudo-inverse alteration into the weight matrix specifically destroys the correlation thereby drastically increasing the memory retrieval efficiency from 0.138 to 0.6 as seen in the above figure. This result effectively orthogonalizes the stored memories thereby reducing interference during retrieval. As a result, the attractor basins remain large over a much wider range of storage loads, allowing the network to successfully correct substantial fractions of flipped bits while maintaining high overlap. Retrieval eventually deteriorates only when the memory load becomes sufficiently large that the finite-size limitations of the network dominate.

4 References

- Dayan, P. and Abbott, L.F. Theoretical Neuroscience: Computational and Mathematical Modeling of Neural Systems.
- Rabinovich, M., Varona, P., Tristan, I., Afraimovich, V. Dynamical Principles in Neuroscience.
- Izhikevich, E.M. Dynamical Systems in Neuroscience.